Unattended audioconferencing

D L Gibson, D Pauley and L Willis

This paper outlines the considerations in designing an audioconferencing algorithm and using it on a commercial speech platform to provide unattended conferencing services.

1. Background and introduction

It is well known that the patent for the telephone instrument was filed by Alexander Graham Bell in February 1876. The potential of the telephone was rapidly seized on by the commercial world in both New England and London, and within two years the first telephone company in the UK was established. The first London exchange opened just three years later with a manual switchboard, and by 1927 it was possible for one 'subscriber' to dial another on the same exchange without any manual intervention.

However, while in these very early days of telephony the operator had the ability to connect several parties together by telephone such that they could all converse, a commercial public service to provide this same capability was not introduced in the UK until the early 1980s with the introduction of BT's 'Rendezvous' service. This service was, and its successors still are, operator based and conference calls generally have to be booked in advance.

The operator establishing a conference call carries out many vital processes, perhaps invisibly, in the creation of the call, e.g. explaining to called parties that an audio-conference is being set up and why, that if lines are busy or engaged further attempts can be made to include them in the conference, and that if the wrong person answers the telephone attempts can be made to find the correct party or suitable messages can be left on the wide variety of answering machines that may be encountered. Any automatic means of establishing conferences needs to be able to address such issues, and it is only within the last few years, with the growing availability of interactive voice response technology, that provision of such an unattended service could be realistically implemented.

This paper looks at one such service, Conference Call 'Instant' (CCI), which is to be shortly introduced by BT. Its implementation can be viewed as a series of three technology layers — the algorithmic layer which provides the raw conferencing capability, the platform layer which provides the environment for the algorithm and connectivity, and the application layer which provides the interaction with its human user. Each of these will now be described in turn.

2. Basic conferencing algorithms

The transmission path from the telephone instrument to the conferencing bridge is shown diagrammatically in Fig 1. The path from a telephone instrument to an exchange is generally an analogue connection. The line card within the exchange splits the incoming speech path from the out-going speech path and will provide any amplification required and filtering previous to converting the speech signal to a digital form ('A'-law PCM in the UK) for onward transmission. Thus a network-based conferencing system in the UK must be capable of digitally combining and distributing multiple streams of 'A'-law coded speech samples. Any practical implementation of such an algorithm must

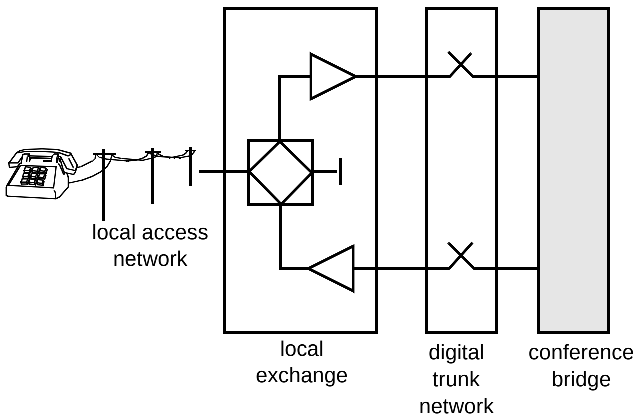


Fig 1 Transmission path.

achieve a satisfactory compromise between digital signal processor (DSP) utilisation, flexibility and subjective speech quality.

A common conferencing algorithm employs the principle of 'principal speaker' selection. This simply monitors the average speech power at each input to the conferencing 'bridge' to determine the principal speaker, and then connects the actual signal from that input to all of the other outputs from the bridge, except the output to the port of the speaker. This technique is very efficient since the input signal is copied directly in 'A'-law form to the output without any conversion. The subjective speech quality of this technique, however, is not ideal, because speakers hear no feedback from the conference participants while talking. They cannot therefore hear the acknow-ledgement signals sent back to them in a normal conversation, such as, 'mmm', 'yes', 'That's right'. This has the effect of speakers feeling they are talking to an 'empty room', and is generally unsettling to unfamiliar users, and may discourage people from using telephone conferencing.

The algorithm employed in the BT Conference Call Instant service employs simple linear mixing of the inputs to the 'bridge', so that each of the participants hears the sum of all of the other inputs except their own input. This means that every participant can hear what anybody else utters at any time. This supplies the 'acknowledgement signals' that provide the normal conversation environment, which people are used to, and find less discomforting than the single speaker approach.

The linear mixing method is less suitable for large conferences when the summation of the background noise sources can become unacceptable, but the CCI service is not intended to cater for larger conferences. In other applications, such as Passepartout [1], the same algorithm can be used but with the addition of voice activity detectors on each input to control the muting of non-speaking inputs. This eliminates the background noise problem while still allowing acknowledgement feedback.

The DSPs used in the speech platform can process 30 speech channels and it would be straightforward to create a 30-channel conferencing bridge in a single DSP, but, if a conference only has five participants, this would achieve very poor utilisation of the valuable DSP resource. The DSP code is therefore arranged to provide five bridges each having six ports. This means that small conferences can be efficiently accommodated, and the bridges can be cascaded when required to accommodate larger conferences. The six DSPs on a single BT Laboratories (BTL) speech processing card used in the speech platform are sufficient to support any mix of conference sizes from a 30-channel line system.

3. Conference bridge platform

The particular platform chosen to host the CCI service is the Millennium-CT (MCT). The system is PC-based and includes two main types of cards. The digital line interface card (DLIC) provides a 30- or a 60-channel connectivity to the telephone network. It contains a non-blocking digital switch to enable digital speech paths to be extended over an industry-standard speech bus (multi-vendor interface protocol) to the signal processing cards. These cards each contain six DSPs along with a controlling Motorola 68000 family processor. The system also contains a (data) network interface card and an alarm card. Finally, the system is controlled by a PC processor and is accom-modated in a single 19-inch rack-mountable chassis. A photograph of the CCI hardware is shown in Fig 2.

Fig 2 Conference Call Instant hardware. The Millennium-CT shelf is the black unit in the right hand rack with its associated monitor above.

The MCT's operating environment is Unix and management/control of the system is vested in a single process. This MCT manager has the capability to run different speech applications on one or more telephone lines, either immediately or on receipt of an incoming call. The manager is also responsible for resource allocation — the assignment of DSPs and lines to particular applications — which it can achieve on a static or dynamic basis.

Once launched by the MCT manager, an application proceeds by requesting the manager to carry out a particular operation such as replaying a message or detecting DTMF tones. At the completion of each operation, the manager will respond to the application with a message indicating the success of the operation along with any ancillary information. Based on the response, the application may then make another request, and then another, and so on until eventually the application terminates.

The CCI service is implemented as one such application launched in response to an incoming phone call. The World Wide Web (WWW) training facility is launched in a similar way when the customer calls a different number. Support for WWW-requested conferences is provided by an application, on the same platform, which the MCT manager launches immediately.

4. Conference Call 'Instant'

The conference calls which people make can probably be split into two types. Firstly, there are those which take place on a regular basis — this might be a weekly sales progress meeting or a monthly management meeting. Secondly, there are those conferences which take place on an *ad hoc* basis, perhaps in response to some event which could not be predicted in advance, such as an act of sabotage. Conference Call Instant can be successfully used in both circumstances. However, it should be remembered that as the size of a conference grows, so also does the amount of effort required to bring together all the various parties, and for a large conference this can become a significant overhead. In such circumstances use of an attended service is to be preferred as the work of collecting together all the conferees can be handled by one or more operators. For this reason the maximum conference size available through Conference Call Instant is limited to ten (which covers more than 90% of all conferences) and distinguishes it from the attended service which can handle more participants and at the same time offer a greater range of facilities, e.g. voting.

4.1 Telephone access

When customers register with the CCI service they will be issued with a customer number (CN), a personal identification number (PIN) and given a telephone number and WWW uniform resource locator (URL) by which access to the service can be gained.

A customer wishing to establish an audioconference using a telephone will ring the given telephone number using a normal DTMF telephone. The call will be routed to an MCT platform which will subsequently act as the conference bridge. The MCT will answer the call, greet the caller and initiate a dialogue to elicit the caller's identity and conference parties to be contacted. The principal design objective for the MCT application program, which provides the CCI telephone service, was that it should be straightforward to use, such that it should not be necessary for the user to constantly refer to a written user guide. This requirement had to be balanced against demands to also provide 'world-beating' sophistication. These aims have been met by allowing users easily to establish straightforward conferences by using only the most basic facilities, and to move on to the more sophisticated features as they become more familiar and more confident in using the system. The application provides full voice guidance at every stage of the conference set-up procedure.

On answering the telephone the CCI application will greet the caller and request them to enter their customer number, followed by their PIN. It will invite new or infrequent users of the system to listen to a spoken description of how to use the service by simply pressing a # when asked for their customer number. After successfully identifying themselves to the system, the user will be asked to enter the number of the person they wish to join their conference. The system will call the given number, and, upon answer, will connect the conference instigator directly to the called party. This is to allow instigators to establish that the wanted person has indeed answered the call, rather than a colleague or an answering service, and to introduce themselves and invite the called party to join the conference. If the correct person has been contacted and is willing and able to join the conference the instigator can instruct the system by pressing a key on his keypad, to accept the called party, and connect them to music-on-hold until all of the other members of the proposed conference have been contacted. At this point instigators will instruct the system to connect all the participants into the conference, to begin their discussions. The conference is terminated by the instigator by simply hanging up the telephone.

4.2 WWW access

The World Wide Web is an alternative means by which conference calls can be established and, for the office user with Internet access, may be a more convenient mechanism. To run the service customers use their browser to access the CCI home page at the URL supplied at registration. The WWW pages have a simple format and can be displayed with all popular browsers. The home page provides links to other pages providing further information about the CCI service and the attended service, the registration process, down-loadable user guides and training facilities, as well as to screens allowing conference calls and personal conference lists (PCL) to be established.

Just as the CCI telephone service catered for both the novice and the experienced user, so also does the WWW system by proving a simple conference set-up as well as an advanced set-up facility. In either case, the essential information to be entered is the same — a customer number, a PIN and the telephone numbers of the conference participants. Assuming a valid CN/PIN combination, the first number in the list is the telephone number which the customer supplied at registration. This number can be edited on the screen as appropriate if, for example, the customer is using the facility while away from his normal place of work.

Other numbers are added in like manner until the list is complete. Clicking on the 'submit' button will result in the conference request being dispatched and a response being displayed on the customer's screen as to whether the request can be currently serviced. The WWW screens play no further part in the conference establishment.

The first stage of conference establishment is that the bridge will ring the chairman who will be expecting the call. If the telephone is engaged or unanswered, the CCI system will subsequently make a number of further attempts and if these are unsuccessful the conference request will be discarded. After announcing the service, the first stage in the dialogue is to ask the chairman to press a key on the telephone keypad as confirmation that the call has not been answered by an answering machine. Upon acknowledgement, the system will then attempt to contact each of the conference parties listed, in the same way as the telephone-based service, with the chairman being asked to accept or reject lines in the usual way with accepted parties being given music-on-hold while the conference set-up progresses. Once all numbers in the list have been contacted the conference is automatically started with the number of other participants being announced. The chairman can subsequently control the conference composition by accessing a special 'options menu'. The conference terminates when the chairman replaces his handset and a customer data record (CDR) is generated for billing purposes.

4.3 Personal conference lists (PCLs)

One of the more sophisticated features of the application is the provision of personal conference lists. It can be quite arduous to key in more than two or three full-length telephone numbers to set up a conference. Also the user may feel under pressure once they have some parties connected to their conference on hold waiting for the conference to begin, and consequently more likely to mis-key the number entries. The provision of PCLs allows users to prepare in advance lists of telephone numbers of parties they wish to join in a conference, and then specify the list they wish the system to use by simply entering a single digit. It is also often the case that customers regularly wish to establish conference calls to the same groups of people and the use of PCLs can speed this process. These lists can be prepared 'off-line', i.e. outside of an actual conference set-up using the telephone or WWW browser. Users can check and correct each number entry at their own pace and thereby avoid any pressure which they may have otherwise experienced if they had had to enter each number during the conference set-up. If a user sets up a conference by keying in on a telephone the numbers during the conference set-up, the user can, at the conclusion of the conference, instruct the system to store in a PCL the numbers of the parties which were connected. Users can then simply instruct the system to implement that PCL the next time they wish to confer with the same parties.

4.4 WWW training

While a design aim has been to make the CCI as simple and as intuitive to use as possible, some aspects of the advanced WWW conference set-up may, perhaps, not be completely apparent to people who have had little exposure to the WWW. To meet this need a WWW training facility has been introduced to guide the customer through PCL and conference creation. To invoke the facility the customer should 'click' on the appropriate link on the CCI home page. The customer will then be advised of a telephone number to ring. Once the call is connected the MCT instructs the training application on what pages to display depending on the user's DTMF response to audio instructions. Audio explanation of the operation of the various WWW pages is given over the telephone and relevant areas of each page are highlighted. The training is terminated by the user selecting a particular option on the telephone or by simply hanging up. It should be noted that this facility calls for the use of 'server push', a feature which is not supported by all browsers.

4.5 Service administration

4.5.1 Registration and customer support

Before using CCI a customer must first contact BT to register and thereby gain access to the service. Such contact may come by referral through operator services or as a result of accessing the CCI WWW home page, itself indirectly linked to BT's public home page. In either case the customer will be asked to provide contact information for billing purposes. The customer is also asked to provide his normal telephone number to be used when establishing conference calls using the WWW facility. Once satisfied on these points all billing details are entered on to the main database and a CN/PIN generated. A 'welcome' pack is then sent advising the customer of the CN/PIN, the URL, the telephone number for the service, and the contact number of an operator for further assistance.

A number of special WWW screens are available to assist the service operator to administer the service. These screens are necessarily username/password protected and the system manager can control the number of administrators able to access customer information. Using these screens an administrator can access the database to identify and update CN and PIN information and thereby access other customer data, such as PCL content. The use of WWW screens enables the systems administration not to be tied to a particular type of equipment or location, giving the service operator greater freedom in how the service is administered.

4.5.2 Billing

At the conclusion of a conference a CDR is produced by the Millennium platform and held locally. These CDRs provide information concerning the CN, bridge, channels used, method of conference establishment, telephone numbers contacted and call duration. These CDRs are periodically transferred to the billing system where they are used for bill production. This system uses a complex algorithm and charging tables, to establish the price of a given conference call. The charge is based on the cost of the outgoing calls together with a conferencing charge for use of the MCT.

4.5.3 MCT operation and configuration

The telephony aspect of the CCI service is provided by multiple MCTs connected to the PSTN via ISDN-30 links. Each MCT has a maximum capacity of 60 channels. A single telephone number is associated with channels on these trunks in such a way as to distribute inbound traffic equally across all MCTs, exercising all available time-slots. The MCT will associate calls on each of these time-slots with the CCI telephone access application or the training application, neither of which is invoked until call arrival. Other time-slots in a trunk are associated by the MCT with the WWW application which is invoked on system start-up. The relationship of application to channel is held in a single table within the MCT and can be readily changed by the MCT administrator, but naturally has to align with the channel/telephone number association provided at the public exchange. The number of channels associated with each application should be determined from traffic estimates/measurements; initially, from each 30-channel group, five will be available for WWW conference creation and two for WWW training with the balance available for conferences to be set up by telephone.

In the event of catastrophic MCT failure, a hardware relay will operate which can be used to raise an alarm. Such a situation might arise due to mains power or trunk loss. In the case of less severe errors, where there is redundancy (such as an individual DSP failure or entire signal processing card failure), the MCT system attempts to reconfigure itself to ensure continuity of service, though perhaps at reduced capacity, and keeps a log of all errors for subsequent inspection. Thus the internal configuration of resources is dynamic, but can be controlled and influenced by the MCT administrator.

MCT administration may be either via a local console or remote via the data network. Generally, there is little need for local access to the system and system maintenance, and support can be provided at a distance.

4.6 Service enhancements

The above CCI service meets the essential requirement of establishing conference calls on demand. However, there are a number of additional features, listed below, which are under consideration for introduction to widen the appeal and usefulness of the service.

- Dial-in conference joining

  The need for this arises for many reasons. Firstly, it may be that a chairman has left a message for a would-be participant who was unavailable at the time the conference call was being established. If a conference dial-in facility existed, the participant could then join the conference when available. Secondly, a conference participant using a mobile telephone might lose telephone contact and with such a facility could rejoin the conference when conditions prevail. A third reason might be that the meeting had been prearranged but some participants could not specify in advance a number on which they could be contacted. With a conference dial-in facility they could join the conference at the previously agreed time. Finally, a fourth reason might simply be to spread the cost of the conference call!

- Conference recording

  A conference call will usually be made with the intent of discussing particular matters, and it may be desirable that a transcript of the discussion be later produced. Thus a facility whereby the chairman can record some or all of the conference discussion for subsequent replay is desirable.

- Use of speech recognition

  While selection of PCLs within CCI is easy via the WWW because they can have names, its use on the telephone is dependent on the user being able to remember the number of the particular PCL that should be used. This problem could be overcome if speech recognition technology were used instead — users would simply have to speak the name of the conference group they wished to use, e.g. ‘sales meeting’, ‘project review meeting’. Speech recognition could probably also be successfully used as an alternative method of responding to certain prompts.

In providing any or all of these enhancements great care needs to be taken to retain the ease of use of the system while ensuring the security and reliability of the service.

5. Audio- and data-conferencing

The purpose of a conference call is generally to discuss some particular issue, and in conducting the discussion participants will often have relevant information at their disposal, perhaps circulated beforehand. At present, this information will generally be in paper form and may be personal to each participant. The usefulness of the meeting will often be significantly improved were all participants able to view data pertinent to the discussion, and may be improved even further if this data could also be manipulated.

Conference Call Instant goes some way towards making this possible. Once a conference call is established, participants with WWW access could view a common page. If one participant (perhaps the chairman) could publish to this known URL others could also view it and see updates, provided they used their browser's refresh option periodically. While such an approach to combined audio/ data conferencing may be acceptable in certain circumstances, better data conferencing tools based on the T.120 standard which allow a richer interaction are now becoming commercially available. To meet this need a useful extension to CCI would be the ability to optionally initiate a T.120-based data conference to enable such sharing and viewing of electronic documents.

This approach supposes that the audio interaction takes place before the data interaction. However, with the advent of developments such as 'shared spaces' on the Internet, this supposition is no longer true. Thus, if the primary contact with other participants is in a data world, the question arises as to whether the unattended audio bridge would take the same form as in a service such as CCI. The answer is that if the control of the audio/data interaction is vested in the data platform the audio bridge must change to act as a server device, performing its operation in response to requests made by the controlling data platform.

To enable the data platform to successfully establish and control audioconferences a messaging protocol needs to be established between the two platforms. Such a system has been implemented based on using RFC1006 protocol to provide an X.224-like transport service to be established across the TCP connection between the two platforms. Using this protocol, messages can be exchanged between the two systems to control the operation of the audio-conferencing platform. These messages fall into five general categories:

- registration functions — these allow the two platforms to pass the necessary set-up information between themselves to register platforms, conferences and lines,
- telephony functions — these allow a data platform to query a line status, clear calls and make calls,
- bridging functions — these allow mixing, speaker and microphone muting, the mixing functions allowing the complete conference to be mixed, groups within a conference to be mixed together as if they were a conference in their own right, and individual mixing functions for each person,
- support functions — these allow requesting of conference and line status,
- inquiry functions — these allow the data platform to request that the audio platform ask a question of the user, the query being a text string which is converted to speech by the audio bridge; any response required is given in the form of DTMF tones which the audio bridge passes on to the data platform.

Two forms of system software allowing such control from a data platform have been implemented using an MCT platform. The first form uses the same MCT manager as for CCI and as such its implementation currently precludes some operations such as sub-conferencing. The second implementation takes the form of a modified MCT manager and potentially allows a more complete implementation.

This form of the bridge has so far been used in conjunction with data platforms to provide two different services. One such system is RISE/MERLIN [2] which is used to assist in the teaching of English as a foreign language. In this system the data platform is a WWW server/database. The other system is 'Passepartout' [1] which is T.120-based data-conferencing platform which has an iconic representation of conference participants.

6. Future directions

There are many possible additional features to the audio bridges to further enhance the overall conferencing capability. These include such items as using voice activity detectors within the bridge to not only perform a microphone muting operation but also indicate to an attached data platform who is speaking. In data-led applications there is also a perceived need for faster audio-conference establishment by parallel dialling on the bridge instead of the present sequential dialling process. A further enhancement is a facility for participants to signal to the bridge using DTMF tones — this might be used, for example, to flag attention or perhaps vote on an issue.

Linking the conference creation capability to other tools on the corporate desktop, such as a meeting scheduler or messaging facility, would also be a valuable enhancement. An integration with 'Groupware' products such as Lotus Notes is another opportunity.

In the further future, conference establishment and management might be established by a 'personal assistant' rather than by the individual. The assistant might appear as a talking head and accept natural language verbal requests for telephone call or conference establishment, garnering, displaying and manipulating data as commanded.

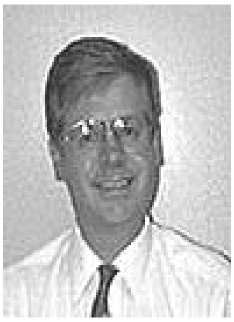
7. Conclusions

The last decade has seen an explosive growth in the audioconferencing business and it is now one of the fastest growing parts of BT's business. The opportunities for continued growth brought about by combining the strengths of telephony, interactive voice response technology and the Internet are tremendous, and the next few years will witness many changes which will affect the way in which people work. Services such as CCI and RISE are only just the beginning; new and broader services will continually emerge. It promises to be an exciting new Millennium!

References

1 Russ M: 'Desktop conversations — the future of multimedia conferencing', BT Technol J, 15, No 4, pp 42—50 (October 1997).

2 Fowler C and Mayes T: 'Applying telepresence to medicine', BT Technol J, 15, No 4, pp 189—196 (October 1997).



David Gibson joined BT in 1974 having obtained a BA degree in Theoretical Physics from Oxford University and an MSc in Telecommunication Systems. He worked for five years at Goonhilly Satellite Earth Station in Cornwall where he was involved in the effect of weather on SHF propagation. On promotion he moved to BT Laboratories in 1979 where he has been involved in Speech Technology. In recent years he has been responsible for development of the Millennium-CT platform and its subsequent use in novel services such as 'Faultsman', 'Portsman' and 'Conference Call Instant'. He is a Member of the IEE and is a Chartered Engineer.

Leslie Willis joined BT Laboratories in September 1996 having obtained a BSc honours degree in Computer Science from the University of Teesside.

In the nine months he has been working for BT he has been involved in a number of projects involving the Internet and speech applications.

He is a member of the British Computer Society (BCS).

Dave Pauley joined BT in 1966 and completed his apprenticeship in the London circuit laboratories where he worked on Strowger exchange enhancements and the development of the 'miniaturised' SSAC9. Upon promotion he then worked on the establishment of regional repair centres for electronic equipment, and the replacement of crossbar common control systems with a control processor.

He transferred to BT Laboratories in 1978 where he worked on the development of custom LSI devices to perform signal processing in high speed modems. He then worked on the definition of the V.32 standard and developed the control software for the world's first production V.32 modem, and later the BT 5th generation multi-mode modem employing DSPs.

Upon transfer to the speech processing unit at Martlesham he worked on the inception of the Minor Applications Platform (MAP), and various algorithm and application developments. He was latterly responsible for the development of the conferencing DSP algorithm, and the conferencing application program used to provide the BT Conference Call Instant service.

He is a Member of the IEE and is a Chartered Engineer.